Online Resource 1
Factor-Adjusted Smoothed Rank Estimation for High-Dimensional Accelerated Failure Time Models

Weineng ZHU · Yang JU

Lifetime Data Analysis
Complete proofs, numerical reproducibility, and implementation details

A Proof of Proposition 2.1

We give the proof for a fixed latent representation. The observed-design statement follows by the same argument, as explained at the end. All conditional expectations in the first three steps are given $(\boldsymbol{D}_1, \boldsymbol{D}_2)$. Write $G_i(t) = G(t \mid \boldsymbol{D}_i)$, $\boldsymbol{w} = \boldsymbol{D}_1 - \boldsymbol{D}_2$, and $\boldsymbol{v} = \boldsymbol{\vartheta} - \boldsymbol{\vartheta}_0$.

Event-weighted differences. On $\{\Delta_i = 1\}$, $R_i = \varepsilon_i$, so the conditional event subdensity is $f_\varepsilon(t)G_i(t)$. Moreover,

$$\Pr(R_j \geq t \mid \boldsymbol{D}_j) = S_\varepsilon(t)G_j(t).$$

It follows that

$$\begin{aligned} A_{12} &:= E\{\Delta_1 \mathbf{1}(R_2 \geq R_1) \mid \boldsymbol{D}_1, \boldsymbol{D}_2\} \\ &= \int f_\varepsilon(t)S_\varepsilon(t)G_1(t)G_2(t)\,\mathrm{d}t = A_{21}. \end{aligned} \tag{A.1}$$

Weineng ZHU (corresponding author) · Yang JU
Department of Biostatistics, City University of Hong Kong, Hong Kong SAR, China
E-mail: weinenzhu2-c@my.cityu.edu.hk

The finite measure $\Pr(\Delta_1 = 1, R_2 - R_1 \in \mathrm{d}r \mid \boldsymbol{D}_1, \boldsymbol{D}_2)$ has a density, say $q_{12}(r)$. With $F_{R_2|\boldsymbol{D}_2}$ denoting the conditional distribution of R_2,

$$q_{12}(r) = \int f_\varepsilon(z - r) G_1(z - r) \, \mathrm{d}F_{R_2|\boldsymbol{D}_2}(z) \le L_\varepsilon \qquad \text{a.e.} \tag{A.2}$$

The Stieltjes integral allows censoring-induced atoms in R_2. Restricting it to the event submeasure of patient 2 gives, for $|r| \le A$,

$$\begin{aligned} q_{12}(r) &\ge \int f_\varepsilon(z - r) G_1(z - r) f_\varepsilon(z) G_2(z) \, \mathrm{d}z \\ &\ge (2A - |r|) m_\varepsilon^2 g_0^2 \ge c_{\mathrm{loc}}, \end{aligned} \tag{A.3}$$

because $I \cap (I + r)$ has length $2A - |r|$. These bounds hold almost everywhere, which suffices for the integrations below.

A finite conditional contrast. Set $t = \boldsymbol{w}^\top \boldsymbol{v}$. At the candidate parameter,

$$e_2^o(\boldsymbol{\vartheta}) - e_1^o(\boldsymbol{\vartheta}) = R_2 - R_1 + t.$$

Define

$$k_{12}(t) = E[\Delta_1 \{(R_2 - R_1 + t)_+ - (R_2 - R_1)_+\} \mid \boldsymbol{D}_1, \boldsymbol{D}_2] \, .$$

This is finite, even without an error moment, because its absolute value is at most $|t|$. The absolute continuity of the event-weighted difference measure gives $k_{12}'(0) = A_{12}$ and $k_{12}''(t) = q_{12}(-t)$ almost everywhere. Integrating this derivative, for $t \ge 0$,

$$k_{12}(t) - A_{12} t = \int_0^t (t - z) q_{12}(-z) \, \mathrm{d}z.$$

For $t < 0$, the corresponding expression is

$$k_{12}(t) - A_{12} t = \int_t^0 (z - t) q_{12}(-z) \, \mathrm{d}z.$$

Nonnegativity of the density and (A.3) imply in either case

$$k_{12}(t) - A_{12} t \ge \frac{c_{\mathrm{loc}}}{2} \rho_A(t). \tag{A.4}$$

For example, when $t > A$ the retained integral is $c_{\mathrm{loc}} \int_0^A (t - z) \, \mathrm{d}z = c_{\mathrm{loc}}(At - A^2/2)$; the case $0 \le t \le A$ gives $c_{\mathrm{loc}} t^2/2$. The negative cases follow analogously.

Cancellation and strict separation. Since the two patients are identically distributed, interchanging their indices preserves their joint law, changes $\boldsymbol{w}$ to $-\boldsymbol{w}$, and leaves A_{12} unchanged by (A.1). The integrability of $\boldsymbol{w}^\top \boldsymbol{v}$ therefore gives

$$E(A_{12}\boldsymbol{w}^\top \boldsymbol{v}) = 0.$$

Taking expectations in (A.4) proves (2.15). When $\boldsymbol{\vartheta}, \boldsymbol{\vartheta}_0 \in \Theta_s$, their difference obeys $\|\boldsymbol{v}_\beta\|_0 \leq 2s$. Assumption 2.3 yields

$$E\{(\boldsymbol{w}^\top \boldsymbol{v})^2\} = 2\,\mathrm{Var}(\boldsymbol{D}_i^\top \boldsymbol{v}) \geq 2\kappa_{\mathrm{pop}}\|\boldsymbol{v}\|_2^2.$$

For nonzero $\boldsymbol{v}$, $\Pr(\boldsymbol{w}^\top \boldsymbol{v} \neq 0) > 0$. Because ρ_A vanishes only at zero, the contrast is strictly positive. Also $\rho_A(t) \leq 2A|t|$, so the expectation in (2.15) is finite. This proves the unique minimum at $\boldsymbol{\vartheta}_0$ on Θ_s without requiring a finite expected uncentered loss.

The ordinary observed-data target. For $\mathcal{M}_0$, set $\boldsymbol{V}_i = (\boldsymbol{Z}_i^\top, \boldsymbol{X}_i^\top)^\top$ and $\boldsymbol{b} = (\boldsymbol{\theta}^\top, \boldsymbol{\beta}^\top)^\top$. Replace $\boldsymbol{w}^\top \boldsymbol{v}$ by $(V_1 - V_2)^\top(\boldsymbol{b} - \boldsymbol{b}_0)$ in the preceding calculation. The event-weighted centering and density bounds can still be established conditional on $(\boldsymbol{D}_1, \boldsymbol{D}_2)$ and then averaged. A positive variance for every nonzero sparse difference in this observed predictor gives strict separation.

Every contrast increment can now be computed as an expectation of a function of the observed pair (O_1, O_2). To avoid an unknown reference parameter in the identification argument, one can fix any deterministic $\boldsymbol{b}_\dagger$ and subtract the loss at $\boldsymbol{b}_\dagger$ instead. Every such contrast is finite and differs from the one centered at $\boldsymbol{b}_0$ by a constant independent of $\boldsymbol{b}$. Thus its unique minimizer is a functional of the observed-data law. The corresponding $\mathcal{M}_1$ result is relative to the specified latent representation. □

B Deterministic properties of the estimator

B.1 Proof of Proposition 3.1

Direct integration of $E(r + hV)_+$ gives (3.6). Jensen's inequality yields $\ell_h(r) \geq r_+$. For $r \geq 0$, the approximation gap $g_h(r) = \ell_h(r) - r$ has derivative $\Phi(r/h) - 1 \leq 0$. The identity $\ell_h(-r) = \ell_h(r) - r$ shows that the gap is even. Its maximum is therefore attained at zero and equals $h\phi(0) = h/\sqrt{2\pi}$. This proves the first inequality in (3.8) without any assumption on observed outcomes.

Symmetry and Fubini's theorem give

$$\begin{aligned}\int_{\mathbb{R}} |\Phi(r/h) - \mathbf{1}\{r \geq 0\}| \, \mathrm{d}r &= 2h \int_0^\infty \{1 - \Phi(t)\} \, \mathrm{d}t \\ &= 2hE(V_+) = h\sqrt{2/\pi}.\end{aligned}$$

The value of the indicator at zero is irrelevant to this integral. Summing the uniform scalar bound uses

$$\frac{1}{n(n-1)} \sum_{i \neq j} \Delta_i = \frac{1}{n} \sum_i \Delta_i = \bar{\Delta},$$

which proves (3.9). Including diagonal terms would add the parameter-independent quantity $n^{-2}h\phi(0) \sum_i \Delta_i$ if the n^{-2} normalization were used, so the stated off-diagonal convention must be retained when comparing objective values.

Differentiating the finite sum gives (3.19), whose Hessian is positive semidefinite. Grouping the coefficients of each $\widehat{\boldsymbol{W}}_i$ proves (3.21); convex subgradient optimality gives (3.20). Differentiation applies directly to the finite empirical sum. Dense gradient evaluation costs $O(nP + n^2)$ arithmetic; pair weights may be blocked to avoid an $N \times P$ matrix.

B.2 Proof of Proposition 3.2 and coordinate identities

The PCA equations imply

$$\boldsymbol{X}_c \boldsymbol{X}_c^\top \widehat{\boldsymbol{F}}/n = \widehat{\boldsymbol{F}}\widehat{\boldsymbol{\Lambda}}, \quad \widehat{\boldsymbol{B}}^\top \widehat{\boldsymbol{B}} = \widehat{\boldsymbol{F}}^\top \boldsymbol{X}_c \boldsymbol{X}_c^\top \widehat{\boldsymbol{F}}/n^2 = \widehat{\boldsymbol{\Lambda}}.$$

Since $\boldsymbol{X}_c \boldsymbol{X}_c^\top \mathbf{1}_n = 0$, selected positive-eigenvalue eigenvectors are centered. The definitions then give $\boldsymbol{J}\widehat{\boldsymbol{U}} = \widehat{\boldsymbol{U}}$, $\widehat{\boldsymbol{F}}^\top \widehat{\boldsymbol{U}} = 0$, and $\boldsymbol{X}_c \widehat{\boldsymbol{B}} = \widehat{\boldsymbol{F}}\widehat{\boldsymbol{\Lambda}}$, proving (3.2).

Substituting (3.13) into the fitted predictor yields

$$\begin{aligned}\widehat{\boldsymbol{W}}\boldsymbol{\vartheta}^{\dagger} &= \boldsymbol{Z}_c\boldsymbol{\theta}_0 + \widehat{\boldsymbol{U}}\boldsymbol{\beta}_0 + \widehat{\boldsymbol{F}}(\widehat{\boldsymbol{B}}^{\top}\boldsymbol{\beta}_0 + \boldsymbol{H}^{-T}\boldsymbol{\xi}_0)\\ &= \boldsymbol{Z}_c\boldsymbol{\theta}_0 + \boldsymbol{X}_c\boldsymbol{\beta}_0 + (\boldsymbol{F}_c\boldsymbol{H}^{\top} + \boldsymbol{E}_F)\boldsymbol{H}^{-T}\boldsymbol{\xi}_0\\ &= \boldsymbol{\eta}_c^0 + \boldsymbol{E}_F\boldsymbol{H}^{-T}\boldsymbol{\xi}_0.\end{aligned}$$

Since $\boldsymbol{J}\boldsymbol{E}_F = \boldsymbol{E}_F$, $\mathbf{d}$ is centered. Subtracting from $Y_i = R_i + \eta_i^0$ gives the residual identities; rowwise Cauchy–Schwarz bounds b_n. Adding and subtracting the reference predictor gives the last decomposition. When $\boldsymbol{\xi}_0 = 0$, the approximation term vanishes without any factor-accuracy assumption.

For orthogonal Q, rotating $\widehat{\boldsymbol{F}}$ and $\widehat{\boldsymbol{B}}$ to $\widehat{\boldsymbol{F}}\boldsymbol{Q}^{\top}$ and $\widehat{\boldsymbol{B}}\boldsymbol{Q}^{\top}$ preserves $\widehat{\boldsymbol{U}}$; taking $\boldsymbol{\gamma}^* = \boldsymbol{Q}\boldsymbol{\gamma}$ preserves the objective. Writing the prediction map as $(\widehat{\boldsymbol{B}}^{\top}\widehat{\boldsymbol{B}})^{-1}\widehat{\boldsymbol{B}}^{\top}\boldsymbol{x}_c$ shows that it rotates to $Q\widehat{\boldsymbol{f}}(\boldsymbol{x})$ and preserves the deployed score, even outside the diagonal PCA eigenbasis.

For a training row $\boldsymbol{x}_{c,i}^{\top}$, the identity $\boldsymbol{X}_c\widehat{\boldsymbol{B}} = \widehat{\boldsymbol{F}}\widehat{\boldsymbol{\Lambda}}$ gives $\widehat{\boldsymbol{\Lambda}}^{-1}\widehat{\boldsymbol{B}}^{\top}\boldsymbol{x}_{c,i} = \widehat{\boldsymbol{f}}_i$. The residual mapping therefore reproduces $\widehat{\boldsymbol{u}}_i$. Finally,

$$\widehat{\boldsymbol{u}}(\boldsymbol{x})^{\top}\widehat{\boldsymbol{\beta}} + \widehat{\boldsymbol{f}}(\boldsymbol{x})^{\top}\widehat{\boldsymbol{\gamma}} = \boldsymbol{x}_c^{\top}\widehat{\boldsymbol{\beta}} + \boldsymbol{x}_c^{\top}\widehat{\boldsymbol{B}}\widehat{\boldsymbol{\Lambda}}^{-1}(\widehat{\boldsymbol{\gamma}} - \widehat{\boldsymbol{B}}^{\top}\widehat{\boldsymbol{\beta}}),$$

which proves (3.17). This proves exact reproduction of the fitted training factors, residuals, and scores.

B.3 Finite attainment and a boundary example

Assume $q > 0$, $h > 0$, $\lambda > 0$, finite observations, and (3.12). On a nonempty sublevel set $Q_{n,h,\lambda}(\boldsymbol{a}, \boldsymbol{\beta}) \le c$, nonnegativity of ℓ_h gives $\|\boldsymbol{\beta}\|_1 \le c/\lambda$. Write

$$t_{ij} = (\widehat{\boldsymbol{A}}_i - \widehat{\boldsymbol{A}}_j)^{\top}\boldsymbol{a}, \qquad z_{ij} = Y_j - Y_i + (\widehat{\boldsymbol{u}}_i - \widehat{\boldsymbol{u}}_j)^{\top}\boldsymbol{\beta}.$$

The inequalities $\ell_h(t + z) \ge (t + z)_+ \ge t_+ - |z|$ imply

$$L_{n,h}(\boldsymbol{a}, \boldsymbol{\beta}) \ge \chi_n\|\boldsymbol{a}\|_2 - C_Y - C_U\|\boldsymbol{\beta}\|_1,$$

where the finite sample constants can be taken as

$$C_Y = N^{-1}\sum_{i\neq j}\Delta_i|Y_j - Y_i|, \qquad C_U = N^{-1}\sum_{i\neq j}\Delta_i\|\widehat{\boldsymbol{u}}_i - \widehat{\boldsymbol{u}}_j\|_{\infty}.$$

Therefore $\|\boldsymbol{a}\|_2 \le \{c + C_Y + C_U c/\lambda\}/\chi_n$ on the sublevel set. It is closed and bounded and hence compact. Taking $c = Q_{n,h,\lambda}(0,0)$ gives a nonempty compact set containing at least one global minimum; the minimizer set is itself compact. If $q = 0$, the penalty directly bounds all coordinates and the same conclusion follows without χ_n.

The sum defining χ_n is nonnegative and continuous on the unit sphere. Thus $\chi_n > 0$ precisely when every nonzero nuisance direction admits a positive event-anchored comparison. The condition concerns the realized event-anchored nuisance design.

For an example of nonattainment, take $n = 2$, $K = 0$, one centered unpenalized covariate with rows -1 and 1, no effective molecular contribution, and $\Delta_1 = 1$, $\Delta_2 = 0$. With scalar coefficient a and $y = Y_2 - Y_1$, the objective reduces to

$$Q(a) = \tfrac{1}{2}\ell_h(y - 2a).$$

It is positive for every finite a, but tends to zero as $a \to \infty$. Thus its infimum is not attained. This remains a valid finite-sample failure mechanism when molecular coefficients are available but optimally zero. Conversely, failure of (3.12) need not imply nonexistence: if all $\Delta_i = 0$, any unpenalized a with $\boldsymbol{\beta} = 0$ is a minimizer, although no slope is learned. Existence, uniqueness, and information for estimation are different requirements.

C Proofs for the conditional theory

Throughout this appendix, E_D and $\Pr_D$ denote expectation and probability conditional on D_n. The design, $\boldsymbol{\vartheta}^\dagger$, h, λ, and the approximation vector $\mathbf{d}$ are fixed. Independent ghost outcomes used below have each patient's own conditional law. The conditional laws are retained separately for each patient.

C.1 Event-weighted differences and the mean score

For each ordered pair let $Q_{ij} = R_j - R_i$, $\zeta_{ij} = d_i - d_j$, and $G_i(t) = G(t \mid \boldsymbol{D}_i)$. On an event observation $R_i = \varepsilon_i$. The conditional event subdensity is $f_\varepsilon(t)G_i(t)$, whereas the conditional survival function of R_j is $S_\varepsilon(t)G_j(t)$. Consequently,

$$A_{ij} := E_D\{\Delta_i \mathbf{1}(Q_{ij} \geq 0)\} = \int f_\varepsilon(t) S_\varepsilon(t) G_i(t) G_j(t)\, \mathrm{d}t = A_{ji}. \tag{C.1}$$

Since $\boldsymbol{w}_{ji} = -\boldsymbol{w}_{ij}$, $N^{-1}\sum_{i \neq j} A_{ij}\boldsymbol{w}_{ij} = 0$. Centering follows from cancellation of the two ordered contributions within each pair.

The finite event-weighted measure $\Pr_D(\Delta_i = 1, Q_{ij} \in \mathrm{d}r)$ has density

$$q_{ij}(r) = \int f_\varepsilon(z - r) G_i(z - r)\, \mathrm{d}F_{R_j \mid \boldsymbol{D}_j}(z) \leq L_\varepsilon \quad \text{a.e.} \tag{C.2}$$

The integral is Stieltjes and permits atoms in R_j. Restricting it to the event submeasure of patient j gives, for $|r| \leq A$,

$$\begin{aligned} q_{ij}(r) &\geq \int f_\varepsilon(z - r) G_i(z - r) f_\varepsilon(z) G_j(z)\, \mathrm{d}z \\ &\geq (2A - |r|) m_\varepsilon^2 g_0^2 \geq c_{\mathrm{loc}}. \end{aligned} \tag{C.3}$$

The intersection of I and $I + r$ has length $2A - |r|$; the location $t_\star$ disappears from the difference interval.

For any real ζ, splitting a shifted smoothed step from an unshifted step yields

$$\int_{\mathbb{R}} |\Phi\{(r + \zeta)/h\} - \mathbf{1}(r \geq 0)|\, \mathrm{d}r \leq |\zeta| + h\sqrt{2/\pi}. \tag{C.4}$$

This follows by adding and subtracting $\mathbf{1}(r + \zeta \geq 0)$, integrating its shift over an interval of length $|\zeta|$, and applying the scalar

smoothing integral from Section 3. The integrated shift bound is linear in $|\zeta|$ with a separate $O(h)$ smoothing term.

Using (C.1)–(C.4), $|\zeta_{ij}| \le 2b$, and Cauchy–Schwarz,

$$|(\boldsymbol{g}_h)_j| \le L_\varepsilon\{2b + h\sqrt{2/\pi}\}\frac{1}{N}\sum_{i\neq l}|w_{il,j}| \le L_\varepsilon\{2b + h\sqrt{2/\pi}\}\sqrt{\widehat{\Gamma}_{jj}} \le 2V_n L_\varepsilon\{2b + h\sqrt{2/\pi}\}.$$

Here $\widehat{\Gamma}_{jj} = 2nV_{n,j}^2/(n-1) \le 4V_n^2$, with $V_{n,j} = \|\widehat{\boldsymbol{W}}_{\cdot j}\|_2/\sqrt{n}$. This proves the mean-score assertion of Lemma 4.1. The same column argument gives $|L_{n,h}(\boldsymbol{\vartheta}^\dagger + \boldsymbol{v}) - L_{n,h}(\boldsymbol{\vartheta}^\dagger)| \le 2V_n\|\boldsymbol{v}\|_1$ and justifies conditional contrast integrability without an error moment.

C.2 A column-scale concentration bound for the gradient

Let G_j be coordinate j of $\nabla L_{n,h}(\boldsymbol{\vartheta}^\dagger)$. Replacing patient i's outcome changes the $2(n-1)$ ordered terms involving that patient. Each scalar weight $\Delta_l\Phi(\cdot)$ lies in $[0,1]$. A valid bounded-difference constant is therefore

$$c_{ij} = \frac{2}{N}\sum_{l\neq i}|w_{il,j}|.$$

By Cauchy–Schwarz and the pairwise variance identity,

$$\sum_i c_{ij}^2 \le \frac{4(n-1)}{N^2}\sum_{i\neq l} w_{il,j}^2 = \frac{4}{n}\widehat{\Gamma}_{jj} \le \frac{16V_n^2}{n}. \tag{C.5}$$

The two-sided bounded-difference inequality consequently gives

$$\Pr_D\{|G_j - E_D G_j| > t\} \le 2\exp\{-nt^2/(8V_n^2)\}.$$

A union bound over P coordinates proves (4.9). This calculation uses patients, not ordered pairs, as the independent units.

C.3 Pairing and moment contraction for the contrast process

For $i < j$ define the symmetric increment

$$H_{ij}(\boldsymbol{v}) = \tfrac{1}{2}\Big[\Delta_i\{\ell_h(r_{ij}^\dagger + \boldsymbol{w}_{ij}^\top\boldsymbol{v}) - \ell_h(r_{ij}^\dagger)\} + \Delta_j\{\ell_h(-r_{ij}^\dagger - \boldsymbol{w}_{ij}^\top\boldsymbol{v}) - \ell_h(-r_{ij}^\dagger)\}\Big], \tag{C.6}$$

where $r_{ij}^{\dagger} = Q_{ij} + \zeta_{ij}$. Extend $H_{ji} = H_{ij}$. For fixed outcomes, $H_{ij}(\boldsymbol{v}) = g_{ij}(\boldsymbol{w}_{ij}^{\top}\boldsymbol{v})$ with $g_{ij}(0) = 0$ and Lipschitz constant at most one. Moreover,

$$L_{n,h}(\boldsymbol{\vartheta}^{\dagger} + \boldsymbol{v}) - L_{n,h}(\boldsymbol{\vartheta}^{\dagger}) = \binom{n}{2}^{-1} \sum_{i<j} H_{ij}(\boldsymbol{v}).$$

Let $m = \lfloor n/2 \rfloor$ and let π be uniform over permutations of the patient labels, independent of the outcomes. Write

$$U_{\pi}(\boldsymbol{v}) = \frac{1}{m} \sum_{l=1}^{m} H_{\pi(2l-1),\pi(2l)}(\boldsymbol{v}).$$

Every unordered pair is equally likely to appear among the m pairs, so

$$E_{\pi} U_{\pi}(\boldsymbol{v}) = \binom{n}{2}^{-1} \sum_{i<j} H_{ij}(\boldsymbol{v}). \tag{C.7}$$

This identity also holds for odd n, where one label is unused in each permutation. Conditional expectations must be averaged over π as well: $E_{\pi} E_D U_{\pi}(\boldsymbol{v}) = \mathcal{R}_h(\boldsymbol{v})$. This average retains the patient-specific conditional laws.

Set $Z(r) = \sup_{\|\boldsymbol{v}\|_1 \le r} |U(\boldsymbol{v}) - E_D U(\boldsymbol{v})|$, where $U(\boldsymbol{v})$ is the left side of (C.7). For any moment order $a \ge 1$, Jensen and Minkowski, followed by ghost-sample symmetrization of the independent pairs, give

$$\begin{aligned} \|Z(r)\|_{L^a(\boldsymbol{D})} &\le E_{\pi} \left\| \sup_{\|\boldsymbol{v}\|_1 \le r} |U_{\pi}(\boldsymbol{v}) - E_D U_{\pi}(\boldsymbol{v})| \right\|_{L^a(\boldsymbol{D})} \\ &\le \frac{2}{m} E_{\pi} \left\| \sup_{\|\boldsymbol{v}\|_1 \le r} \left| \sum_{l=1}^{m} \sigma_l H_{\pi(2l-1),\pi(2l)}(\boldsymbol{v}) \right| \right\|_{L^a(\boldsymbol{D},\sigma)}, \end{aligned} \tag{C.8}$$

where the σ_l are independent Rademacher signs. The maps g_{ij} may depend on outcomes; after these outcomes are fixed, they are deterministic contractions independent of the signs. The absolute-value moment contraction principle (Ledoux and Talagrand, 1991, Chapter 4), with the safe factor two, and ℓ_1/ℓ_∞ duality yield

$$\|Z(r)\|_{L^a(\boldsymbol{D})} \le \frac{4r}{m} E_{\pi} \left\| \left\| \sum_{l=1}^{m} \sigma_l \boldsymbol{w}_{\pi(2l-1),\pi(2l)} \right\|_{\infty} \right\|_{L^a(\sigma)}. \tag{C.9}$$

Suprema are measurable: the finite-dimensional closed ℓ_1 ball is compact, and the functions are continuous in $\boldsymbol{v}$, so a fixed countable dense subset gives the same supremum.

For every pairing and coordinate j,

$$\sum_{l=1}^{m} w_{\pi(2l-1),\pi(2l),j}^2 \le 2\sum_{i=1}^{n} \widehat{W}_{ij}^2 \le 2nV_n^2. \tag{C.10}$$

Consequently the maximum $T_\pi = \|\sum_l \sigma_l \boldsymbol{w}_{\pi(2l-1),\pi(2l)}\|_\infty$ obeys

$$\Pr_\sigma\{T_\pi > B\sqrt{2\log(2P)} + B\sqrt{2t}\} \le \mathrm{e}^{-t}, \qquad B = \sqrt{2n}\, V_n.$$

Integration of this tail gives

$$\|T_\pi\|_{L^a(\sigma)} \le B\{\sqrt{2\log(2P)} + \sqrt{2a}\} \le 2B\sqrt{\log(2P) + a}. \tag{C.11}$$

For example, the positive part above the first threshold has ath moment at most $(\sqrt{2}B)^a \Gamma(a/2+1)$; the elementary bound $\Gamma(a/2+1)^{1/a} \le \sqrt{a}$ for $a \ge 1$ gives the displayed inequality. Since $m \ge n/3$ for $n \ge 2$, equations (C.9)–(C.11) imply

$$\|Z(r)\|_{L^a(\boldsymbol{D})} \le 24\sqrt{2}\, V_n r\sqrt{\{\log(2P) + a\}/n}. \tag{C.12}$$

Take $a = \log(1/\alpha) > 1$. Markov's inequality at $e\|Z(r)\|_{L^a(\boldsymbol{D})}$ gives failure probability at most $\mathrm{e}^{-a} = \alpha$. Because $24\sqrt{2}e < 96$, this proves (4.10). The proof uses the standard pairing strategy for U-processes (Clémençon et al., 2008), but keeps conditional nonidentical laws and the column norm explicit. Bounding the supremum directly by the largest coordinate would lose this refinement.

C.4 Proof of Lemma 4.2

Differentiation of the conditional contrast is valid for fixed $h > 0$: first and second derivatives of each loss increment are bounded by fixed multiples of $\|\boldsymbol{w}_{ij}\|$ and $\|\boldsymbol{w}_{ij}\|^2/h$. For any real shift z, the scalar second derivative is

$$E_D\{\Delta_i \ell_h''(Q_{ij} + z)\} = \int h^{-1}\phi\{(r+z)/h\} q_{ij}(r)\, \mathrm{d}r. \tag{C.13}$$

If $|z| \le A/2$ and $h \le A/2$, the interval $[-z-h, -z+h]$ is contained in $[-A, A]$. Restricting the integral to that interval and using (C.3)

gives the lower bound $c_{\mathrm{loc}}\{2\Phi(1)-1\} = c_{\mathrm{loc}}c_\phi$. For $t \in [0,1]$ the shift along the parameter segment is

$$z = \zeta_{ij} + t\boldsymbol{w}_{ij}^\top \boldsymbol{v}, \qquad |z| \le 2b + 2M_n\|\boldsymbol{v}\|_1.$$

Thus (4.11) keeps the entire segment in the above interval. Integrating the second derivative along this segment gives

$$\mathcal{R}_h(\boldsymbol{v}) - \boldsymbol{g}_h^\top \boldsymbol{v} \ge \frac{c_{\mathrm{loc}}c_\phi}{2N}\sum_{i\neq j}(\boldsymbol{w}_{ij}^\top \boldsymbol{v})^2.$$

The conditional restricted-design inequality proves the result. The factor c_ϕ comes from retaining one bandwidth on each side of a kernel center; it is not an error-density assumption.

C.5 Localization, finite minimizers, and error bounds

We prove the general version with approximation bound b, which also proves the first part of Corollary 4.1. Put $\mu = \mu(b,h)$ and assume

$$\lambda \ge 4\mu + 384V_n t_\alpha,$$
$$R = 8\sqrt{k}\lambda/\kappa,$$
$$h \le A/2,$$
$$2b + 8M_n\sqrt{k}R \le A/2.$$

Use Lemma 4.1 with the fixed conditional radius $r = 4\sqrt{k}R$. On the intersection of its two events,

$$\|\nabla L_{n,h}(\boldsymbol{\vartheta}^\dagger)\|_\infty \le \mu + \sqrt{8}V_n t_\alpha \le \lambda/2.$$

Suppose $\boldsymbol{v}$ satisfies $Q(\boldsymbol{\vartheta}^\dagger + \boldsymbol{v}) \le Q(\boldsymbol{\vartheta}^\dagger)$. Convexity of $L_{n,h}$ and the support of $\boldsymbol{\beta}_0$ imply

$$\begin{aligned} 0 &\ge Q(\boldsymbol{\vartheta}^\dagger + \boldsymbol{v}) - Q(\boldsymbol{\vartheta}^\dagger) \\ &\ge -\tfrac{\lambda}{2}\|\boldsymbol{v}\|_1 + \lambda\{\|(\boldsymbol{v}_\beta)_{S^c}\|_1 - \|(\boldsymbol{v}_\beta)_S\|_1\}. \end{aligned}$$

Rearrangement proves (4.16), and therefore $\boldsymbol{v} \in \mathcal{C}(S)$ and $\|\boldsymbol{v}\|_1 \le 4\sqrt{k}\|\boldsymbol{v}\|_2$.

Now assume additionally that $\|\boldsymbol{v}\|_2 = R$. The local curvature condition holds, and the risk contrast, empirical contrast bound, and penalty inequality give

$$\begin{aligned} Q(\boldsymbol{\vartheta}^\dagger + \boldsymbol{v}) - Q(\boldsymbol{\vartheta}^\dagger) &\geq \tfrac{\kappa}{2}R^2 - \mu\|\boldsymbol{v}\|_1 - 96V_n r t_\alpha - \lambda\|(\boldsymbol{v}_\beta)_S\|_1 \\ &\geq \tfrac{\kappa}{2}R^2 - \{4\mu + 384V_n t_\alpha + \lambda\}\sqrt{k}R \\ &\geq \tfrac{\kappa}{2}R^2 - 2\lambda\sqrt{k}R = \tfrac{\kappa}{4}R^2 > 0. \end{aligned} \tag{C.14}$$

This contradicts the assumed sublevel inequality. Thus no point in the sublevel set through $\boldsymbol{\vartheta}^\dagger$ lies on the sphere of radius R.

If a point of that sublevel set were outside the ball, its line segment to $\boldsymbol{\vartheta}^\dagger$, by convexity of Q, would give a point on the excluded sphere. Hence the whole sublevel set is inside the ball. It is nonempty and closed, so it is compact. The continuous objective attains a minimum there, and every point outside the sublevel set has a strictly larger value than $Q(\boldsymbol{\vartheta}^\dagger)$. This minimum is therefore global. All global minimizers lie in that compact set and satisfy

$$\|\widehat{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}^\dagger\|_2 \leq R, \qquad \|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_1 \leq 4\sqrt{k}R.$$

The set of minimizers is closed and compact as well. Its nonemptiness is proved rather than assumed. Substituting R gives (4.15); the event has conditional probability at least $1 - 2\alpha$. Setting $b = 0$ proves Theorem 4.1. Using the representation (4.28) proves the conditional extension.

D Primitive factor-design guarantees

Constants $C, c > 0$ in this appendix can change between displays and depend only on the fixed constants of Assumption 4.2 and on d, K. Dependence on the outcome constants is added only in the final estimation step. The probability parameter is δ, and $L_\delta, \tau_\delta, a_\delta, r_{F,\delta}$ are defined in (4.19).

D.1 One joint covariate event

Let $\boldsymbol{D}$ be the matrix with rows $\boldsymbol{D}_i^\top$ and $\boldsymbol{D}_c = \boldsymbol{J}\boldsymbol{D}$. We first construct an event of probability at least $1-\delta$ on which

$$\|\boldsymbol{D}_c^\top \boldsymbol{D}_c/n - \boldsymbol{\Sigma}_D\|_{\max} \le C\tau_\delta, \qquad \max_{i,j} |D_{ij}| \le C\sqrt{L_\delta}, \tag{D.1}$$

$$\|\boldsymbol{U}_c\|_{\mathrm{op}} \le C(\sqrt{n} + \sqrt{p}), \qquad \max_i \|\boldsymbol{u}_{c,i}\|_2 \le C\sqrt{p}, \tag{D.2}$$

$$\max_i \|\boldsymbol{B}^\top \boldsymbol{u}_{c,i}\|_2 \le C\sqrt{pL_\delta}, \qquad c\boldsymbol{I}_K \preceq \boldsymbol{F}_c^\top \boldsymbol{F}_c/n \preceq C\boldsymbol{I}_K. \tag{D.3}$$

Here $\boldsymbol{u}_{c,i}$ is row i of $\boldsymbol{U}_c$ written as a column. The prerequisite is $n \wedge p \ge CL_\delta$. The following concentration bounds construct this joint design event.

For every coordinate j, sub-Gaussian tails and a union bound over nP variables give the maximum in (D.1). The sample means obey $\max_j |\bar{\boldsymbol{D}}_j| \le C\sqrt{L_\delta/n}$. Products of two uniformly sub-Gaussian coordinates are uniformly sub-exponential; this does not require independence between the coordinates. Bernstein's inequality and a union bound over P^2 products give

$$\max_{j,l} |n^{-1} \sum_i D_{ij} D_{il} - \boldsymbol{\Sigma}_{D,jl}| \le C\{\sqrt{L_\delta/n} + L_\delta/n\}.$$

Subtracting $\bar{\boldsymbol{D}}\bar{\boldsymbol{D}}^\top$ proves the covariance part of (D.1). Since K is fixed and $\boldsymbol{\Sigma}_f = \boldsymbol{I}_K$, its factor block gives the final spectral statement of (D.3) when the size constant is sufficiently large. It also gives

$$\max_j \|\boldsymbol{U}_{c,\cdot j}\|_2/\sqrt{n} \le C, \qquad \max_j \|\boldsymbol{F}_c^\top \boldsymbol{U}_{c,\cdot j}/n\|_2 \le C\tau_\delta, \tag{D.4}$$

because $E(\boldsymbol{f}_i \boldsymbol{u}_i^\top) = 0$.

For completeness, a bilinear net argument controls the noise operator norm even though coordinates within each $\boldsymbol{u}_i$ may be dependent. For fixed unit vectors $\boldsymbol{a} \in \mathbb{R}^n$ and $b \in \mathbb{R}^p$, $\boldsymbol{a}^\top \boldsymbol{U} b = \sum_i a_i b^\top \boldsymbol{u}_i$ is centered sub-Gaussian with a fixed norm, by independence across patients and the joint sub-Gaussian condition. Apply its tail bound to 1/4-nets of the two spheres, with at most 9^n and 9^p points. The net approximation and a union bound yield

$$\|\boldsymbol{U}\|_{\mathrm{op}} \le C\{\sqrt{n} + \sqrt{p} + \sqrt{\log(1/\delta)}\}.$$

Centering is a contraction, and $n \wedge p \ge C L_\delta$ absorbs the last term. A 1/2-net for a single row, followed by a union bound over patients, gives $\max_i \|\boldsymbol{u}_i\|_2 \le C(\sqrt{p} + \sqrt{L_\delta})$. Since $\|\bar{\boldsymbol{u}}\|_2 \le \max_i \|\boldsymbol{u}_i\|_2$, centering costs at most a factor two, proving (D.2). These are standard net and concentration tools for jointly sub-Gaussian vectors (Vershynin, 2012).

Each coordinate of $\boldsymbol{B}^\top \boldsymbol{u}_i$ has sub-Gaussian norm at most $C\sqrt{p}$, because $\|\boldsymbol{B}_{\cdot l}\|_2 \le C\sqrt{p}$. A union bound over the fixed K projections and n patients, with the same centering argument, proves the first part of (D.3). Assigning the constituent events fixed fractions of δ and increasing C makes their intersection have probability at least $1 - \delta$. All statements below are deterministic on this intersection. In particular, $\boldsymbol{H}$ may be a function of the same data; the bounds are uniform over the relevant vectors and do not require $\boldsymbol{H}$ to be independent.

D.2 Spectral separation and an explicit alignment

The signal matrix $\boldsymbol{F}_c \boldsymbol{B}^\top$ has rank K and nonzero singular values between $c\sqrt{np}$ and $C\sqrt{np}$. The perturbation $\boldsymbol{U}_c$ has norm at most $C(\sqrt{n} + \sqrt{p})$. The singular-value perturbation inequality therefore implies

$$c\sqrt{np} \le s_K(\boldsymbol{X}_c) \le s_1(\boldsymbol{X}_c) \le C\sqrt{np}, \qquad s_{K+1}(\boldsymbol{X}_c) \le \|\boldsymbol{U}_c\|_{\mathrm{op}},$$

where the last singular value is read as zero if the index exceeds the dimension. With sufficiently large $n \wedge p$, the latter is strictly below $s_K(\boldsymbol{X}_c)$. Thus $cp\boldsymbol{I}_K \preceq \widehat{\boldsymbol{\Lambda}} \preceq Cp\boldsymbol{I}_K$ and there is a gap at the selected rank.

The PCA identities give, without any probabilistic approximation,

$$\widehat{\boldsymbol{F}} = \boldsymbol{X}_c\widehat{\boldsymbol{B}}\widehat{\boldsymbol{\Lambda}}^{-1} = \boldsymbol{F}_c\boldsymbol{B}^\top\widehat{\boldsymbol{B}}\widehat{\boldsymbol{\Lambda}}^{-1} + \boldsymbol{U}_c\widehat{\boldsymbol{B}}\widehat{\boldsymbol{\Lambda}}^{-1} = \boldsymbol{F}_c\boldsymbol{H}^\top + \boldsymbol{E}_F,$$

with $\boldsymbol{H}, \boldsymbol{E}_F$ from (4.20). Since $\widehat{\boldsymbol{B}}^\top\widehat{\boldsymbol{B}} = \widehat{\boldsymbol{\Lambda}}$,

$$\|\widehat{\boldsymbol{B}}\widehat{\boldsymbol{\Lambda}}^{-1}\|_{\mathrm{op}} = \lambda_{\min}(\widehat{\boldsymbol{\Lambda}})^{-1/2} \le C/\sqrt{p}.$$

It follows that

$$\frac{\|\boldsymbol{E}_F\|_{\mathrm{op}}}{\sqrt{n}} \le C\left(\frac{1}{\sqrt{n}} + \frac{1}{\sqrt{p}}\right). \tag{D.5}$$

This derivation does not first assume that $\boldsymbol{H}$ is invertible. Instead, $\widehat{\boldsymbol{F}}$ has all singular values $\sqrt{n}$, and (D.5) makes $\boldsymbol{F}_c\boldsymbol{H}^\top = \widehat{\boldsymbol{F}} - \boldsymbol{E}_F$ have all its nonzero singular values bounded above and below by fixed multiples of $\sqrt{n}$. Since $\boldsymbol{F}_c$ has the same scale, $\boldsymbol{H}$ must be invertible, with $\|\boldsymbol{H}\|_{\mathrm{op}} + \|\boldsymbol{H}^{-1}\|_{\mathrm{op}} \le C$. Sample alignment is not assumed orthogonal, consistently with the normalization distinction discussed by Bai and Ng (2013).

D.3 Maximum-row factor error and maximum-column residual error

Set $\boldsymbol{A}_F = \boldsymbol{F}_c^\top\widehat{\boldsymbol{F}}/n$, so $\|\boldsymbol{A}_F\|_{\mathrm{op}} \le C$ and

$$\widehat{\boldsymbol{B}} = \boldsymbol{B}\boldsymbol{A}_F + \boldsymbol{U}_c^\top\widehat{\boldsymbol{F}}/n.$$

For a row error written as a column,

$$e_{F,i} = \widehat{\boldsymbol{\Lambda}}^{-1}\{\boldsymbol{A}_F^\top\boldsymbol{B}^\top\boldsymbol{u}_{c,i} + \widehat{\boldsymbol{F}}^\top\boldsymbol{U}_c\boldsymbol{u}_{c,i}/n\}. \tag{D.6}$$

The first term is at most $C\sqrt{L_\delta/p}$. For the second, apply only deterministic norm inequalities:

$$\begin{aligned}\frac{C}{p}\frac{\|\widehat{\boldsymbol{F}}\|_{\mathrm{op}}}{n}\|\boldsymbol{U}_c\|_{\mathrm{op}}\max_i\|\boldsymbol{u}_{c,i}\|_2 &\le \frac{C}{p}\frac{1}{\sqrt{n}}(\sqrt{n}+\sqrt{p})\sqrt{p}\\ &\le C(p^{-1/2}+n^{-1/2}).\end{aligned}$$

Combining the two proves the maximum-row bound in (4.21). The calculation is valid with same-sample factor estimation: no independence between $\widehat{\boldsymbol{F}}$ and $\boldsymbol{U}_c$ was used.

Let $\widehat{P}_F = \widehat{\boldsymbol{F}}\widehat{\boldsymbol{F}}^\top/n$ and let $\boldsymbol{b}_j$ be column j of $\boldsymbol{B}^\top$. The residual difference satisfies

$$\widehat{\boldsymbol{U}}_{\cdot j} - \boldsymbol{U}_{c,\cdot j} = (I - \widehat{P}_F)\boldsymbol{F}_c\boldsymbol{b}_j - \widehat{P}_F\boldsymbol{U}_{c,\cdot j}. \tag{D.7}$$

The first term equals $-(I - \widehat{P}_F)\boldsymbol{E}_F\boldsymbol{H}^{-T}\boldsymbol{b}_j$ and, divided by $\sqrt{n}$, is at most $C(n^{-1/2} + p^{-1/2})$, because $\|\boldsymbol{b}_j\|_2 \le L_B$. For the second term,

$$\begin{aligned}\frac{\|\widehat{P}_F\boldsymbol{U}_{c,\cdot j}\|_2}{\sqrt{n}} &= \left\|\frac{\widehat{\boldsymbol{F}}^\top\boldsymbol{U}_{c,\cdot j}}{n}\right\|_2 \\ &\le \|\boldsymbol{H}\|_{\mathrm{op}}\left\|\frac{\boldsymbol{F}_c^\top\boldsymbol{U}_{c,\cdot j}}{n}\right\|_2 + \frac{\|\boldsymbol{E}_F\|_{\mathrm{op}}}{\sqrt{n}}\frac{\|\boldsymbol{U}_{c,\cdot j}\|_2}{\sqrt{n}} \\ &\le C\{\tau_\delta + p^{-1/2}\}.\end{aligned}$$

Here $n^{-1/2} \le \tau_\delta$. This proves the maximum-column residual bound in (4.22). Factor-error operator control and residual column control imply

$$\begin{gathered}\max_{j\le P}\frac{\|\widehat{\boldsymbol{W}}_{\cdot j} - \boldsymbol{W}_{\cdot j}\|_2}{\sqrt{n}} \le Ca_\delta, \\ \boldsymbol{W} = (\boldsymbol{Z}_c, \boldsymbol{F}_c\boldsymbol{H}^\top, \boldsymbol{U}_c).\end{gathered} \tag{D.8}$$

The clinical block has zero approximation error.

To control the scales in the conditional theorem, first note

$$\max_j \|\boldsymbol{X}_{c,\cdot j}\|_2/\sqrt{n} \le \max_j\{\|\boldsymbol{F}_c\|_{\mathrm{op}}\|\boldsymbol{b}_j\|_2/\sqrt{n} + \|\boldsymbol{U}_{c,\cdot j}\|_2/\sqrt{n}\} \le C.$$

Since $\widehat{\boldsymbol{U}} = (I - \widehat{P}_F)\boldsymbol{X}_c$, its column norms are no larger. The factor columns have norm $\sqrt{n}$, and clinical columns are bounded using (D.1). Thus $V_n \le C$. Also

$$\max_j \|\widehat{\boldsymbol{B}}_{j\cdot}\|_2 \le \max_j\{\|\boldsymbol{b}_j\|_2\|\boldsymbol{A}_F\|_{\mathrm{op}} + \|\boldsymbol{U}_{c,\cdot j}\|_2/\sqrt{n}\} \le C.$$

Using $\widehat{\boldsymbol{F}} = \boldsymbol{F}_c\boldsymbol{H}^\top + \boldsymbol{E}_F$ and the maximum-row bound gives $\max_i \|\hat{\boldsymbol{f}}_i\|_2 \le C\sqrt{L_\delta}$. Together with $\widehat{\boldsymbol{U}} = \boldsymbol{X}_c - \widehat{\boldsymbol{F}}\widehat{\boldsymbol{B}}^\top$ and the row-loading bound, this yields $M_n \le C\sqrt{L_\delta}$. If the raw $\boldsymbol{D}_i$ coordinates are uniformly bounded, the centered $\boldsymbol{F}_c, \boldsymbol{U}_c, \boldsymbol{Z}_c$ coordinates are bounded, while $r_{F,\delta}$ is bounded under $n \wedge p \ge CL_\delta$. The same argument then gives $M_n \le C$. Coordinate boundedness is an additional assumption, not a consequence of observing a finite data matrix.

D.4 Ideal geometry and proof of the transfer lemma

Write $\boldsymbol{W} = \boldsymbol{D}_c\boldsymbol{T}_H$ with $\boldsymbol{T}_H = \mathrm{diag}(\boldsymbol{I}_d, \boldsymbol{H}^\top, \boldsymbol{I}_p)$. On the joint event, $\|\boldsymbol{T}_H\boldsymbol{v}\|_2 \geq c\|\boldsymbol{v}\|_2$ and $\|\boldsymbol{T}_H\boldsymbol{v}\|_1 \leq C\|\boldsymbol{v}\|_1$ because K is fixed and $\boldsymbol{H}, \boldsymbol{H}^{-1}$ are controlled. These inequalities hold for the data-dependent $\boldsymbol{T}_H$ pointwise. The covariance event gives

$$\begin{aligned}\|\boldsymbol{W}\boldsymbol{v}\|_2^2/n &\geq (\boldsymbol{T}_H\boldsymbol{v})^\top \boldsymbol{\Sigma}_D(\boldsymbol{T}_H\boldsymbol{v}) - C\tau_\delta\|\boldsymbol{T}_H\boldsymbol{v}\|_1^2 \\ &\geq (c_0 - C_0 k\tau_\delta)\|\boldsymbol{v}\|_2^2, \qquad \boldsymbol{v} \in \mathcal{C}(S). \end{aligned} \tag{D.9}$$

For $k\tau_\delta$ sufficiently small this is bounded below by a fixed multiple of $\|\boldsymbol{v}\|_2^2$. Centering yields $\|\boldsymbol{W}\boldsymbol{v}\|_{\mathrm{pair}}^2 = 2\|\boldsymbol{W}\boldsymbol{v}\|_2^2/(n-1)$, so the ideal pairwise restricted bound holds with some fixed $\kappa_0 > 0$.

For any vector $\boldsymbol{x} \in \mathbb{R}^n$,

$$\|\boldsymbol{x}\|_{\mathrm{pair}}^2 = \frac{2}{n-1}\|\boldsymbol{J}\boldsymbol{x}\|_2^2. \tag{D.10}$$

Under the premise of Lemma 4.3,

$$\|(\widehat{\boldsymbol{W}} - \boldsymbol{W})\boldsymbol{v}\|_2/\sqrt{n} \leq \sum_j |v_j|\|\widehat{\boldsymbol{W}}_{\cdot j} - \boldsymbol{W}_{\cdot j}\|_2/\sqrt{n} \leq a\|\boldsymbol{v}\|_1.$$

Equation (D.10) and $n \geq 2$ give

$$\|(\widehat{\boldsymbol{W}} - \boldsymbol{W})\boldsymbol{v}\|_{\mathrm{pair}} \leq 2a\|\boldsymbol{v}\|_1 \leq 8a\sqrt{k}\|\boldsymbol{v}\|_2 \quad (\boldsymbol{v} \in \mathcal{C}(S)).$$

The triangle inequality proves (4.24). Combine it with (D.8) and (D.9). If $\sqrt{k}a_\delta$ is sufficiently small, the resulting lower bound is $\kappa_X \geq c_X > 0$. This completes Proposition 4.1. The max-norm covariance argument yields the sufficient requirement on $k\tau_\delta$.

D.5 Integrating the conditional result

On $\mathcal{E}_D$ in the ordinary model, $b_n = 0$, $V_n \leq C$, $M_n \leq C\sqrt{L_\delta}$, and the curvature constant has a fixed positive lower bound. Apply Theorem 4.1 with $\alpha = \delta$. Since $t_\alpha \leq t_\delta$ and $\mu(0,h) \leq Ch$, sufficiently large C_* in (4.25) makes the conditional penalty inequality hold. The condition $k\sqrt{L_\delta}\lambda \leq c_*$ ensures

$$8M_n\sqrt{k}R = 64M_nk\lambda/\kappa \leq A/2$$

for sufficiently small c_*. In the bounded-coordinate case, $k\lambda \le c_*$ suffices. The conditional bounds become (4.26).

The total failure probability is at most

$$\Pr(\mathcal{E}_D^c) + E\{\mathbf{1}_{\mathcal{E}_D} \Pr(\text{outcome failure} \mid D_n)\} \le \delta + 2\delta = 3\delta.$$

No independence between the design event and the outcome event is needed. The theorem therefore controls the feasible estimator, not only a latent-design minimizer.

For Corollary 4.1, $b_n \le \max_i \|e_{F,i}\|_2 \|\boldsymbol{H}^{-T}\boldsymbol{\xi}_0\|_2 \le C\Xi r_{F,\delta}$ on $\mathcal{E}_D$. The penalty level in (4.30) dominates $4\mu(b_n, h) + 384V_n t_\alpha$. Its locality inequality ensures $2b_n + 64M_n k\lambda/\kappa \le A/2$. The conditional extension proved in Appendix C.5, followed by the same probability integration, gives (4.31).

Finally, with $p \ge n$ and $h \le Ct_\delta$, we have $\tau_\delta \lesssim t_\delta$ when L_δ/n is small, and $a_\delta \lesssim t_\delta$. The jointly sub-Gaussian condition $k^2 L_\delta^2/n \to 0$ implies both $k\tau_\delta \to 0$ and $\sqrt{k}a_\delta \to 0$, and ensures locality. With bounded coordinates, $k^2 L_\delta/n \to 0$ suffices. For the compatible example in the text, independent Rademacher coordinates have identity covariance and a uniformly bounded joint sub-Gaussian norm by the product moment generating function. The Cauchy density is globally bounded and is at least $1/(2\pi)$ on $[-1, 1]$; $C_i^* = 2$ gives $G_i = 1$ there. Positive censoring probability is $\Pr(\varepsilon_i > 2) > 0$. Taking $p = n^2$, $k = \lfloor n^{1/4} \rfloor + d + 1$, and $\delta_n = n^{-2}$ makes $k^2 L_{\delta_n}/n \to 0$. This verifies compatibility without a finite error mean or variance.

E Computational identities and convergence proofs

E.1 Pairwise derivatives, Laplacian bound, and implicit products

Let $\boldsymbol{w}_{ij} = \widehat{\boldsymbol{W}}_i - \widehat{\boldsymbol{W}}_j$. Since $\partial r_{ij}/\partial \boldsymbol{\vartheta} = \boldsymbol{w}_{ij}$, differentiation of a finite sum gives

$$\nabla f_h(\boldsymbol{\vartheta}) = \frac{1}{N} \sum_{i \neq j} \Delta_i \Phi(r_{ij}/h) \boldsymbol{w}_{ij},$$

$$\nabla^2 f_h(\boldsymbol{\vartheta}) = \frac{1}{Nh} \sum_{i \neq j} \Delta_i \phi(r_{ij}/h) \boldsymbol{w}_{ij} \boldsymbol{w}_{ij}^\top.$$

Collecting the positive coefficient of row i and the negative coefficient of column i gives (5.2). Summing c_i over i gives zero. Blocking the row sum does not change any term. Only rows with $\Delta_i = 1$ need to be evaluated, but each such row still includes all other patients.

Combining the (i, j) and (j, i) outer products yields

$$\sum_{i \neq j} \Delta_i \boldsymbol{w}_{ij} \boldsymbol{w}_{ij}^\top = \sum_{i<j} (\Delta_i + \Delta_j) \boldsymbol{w}_{ij} \boldsymbol{w}_{ij}^\top = \widehat{\boldsymbol{W}}^\top \mathscr{L}_\Delta \widehat{\boldsymbol{W}}.$$

The diagonal of this Laplacian is $(n-2)\Delta_i + m$, and its off-diagonal entry is $-(\Delta_i + \Delta_j)$, proving (5.4). The bound $\phi \leq 1/\sqrt{2\pi}$ then proves (5.5), with the Lipschitz property obtained by integrating the Hessian along the segment between two finite points. Centering removes both terms involving $\widehat{\boldsymbol{W}}^\top \mathbf{1}_n$. The operator-norm bound follows from $0 \preceq \operatorname{diag}(\boldsymbol{\Delta}) \preceq \boldsymbol{I}_n$.

For the positive line-search certificate, Taylor's integral remainder is

$$\begin{aligned} f_h(\boldsymbol{y}) - f_h(\boldsymbol{x}) - \nabla f_h(\boldsymbol{x})^\top (\boldsymbol{y} - \boldsymbol{x}) \\ = \frac{1}{Nh} \sum_{i \neq j} \Delta_i d_{ij}^2 \int_0^1 (1 - s) \phi\{(r_{ij}(\boldsymbol{x}) + s d_{ij})/h\} \, \mathrm{d}s. \end{aligned}$$

The normal density decreases as its argument's absolute value increases. Its maximum on the segment is therefore $\phi(m_{ij}/h)$. Bounding the integral by half that maximum proves the first inequality in (5.8); bounding it by $\phi(0)/2$ and using the Laplacian proves the second. Thus this alternative test also accepts every $t \leq \omega/\mathscr{B}_h$, which is the only threshold property needed in the convergence proof.

For completeness, Hessian-vector products can be computed without forming the Hessian. Let $t_i = \widehat{\boldsymbol{W}}_i^\top \boldsymbol{v}$ and, for $i \neq j$, define

$$R_{ij} = \frac{\Delta_i}{h}\phi(r_{ij}/h)(t_i - t_j), \qquad R_{ii} = 0, \qquad b_i = \sum_j R_{ij} - \sum_j R_{ji}.$$

Then $\nabla^2 f_h(\boldsymbol{\vartheta})\boldsymbol{v} = \widehat{\boldsymbol{W}}^\top b/N$. This formula is useful for testing, although Algorithm 1 does not require a Hessian-vector product. Substituting $\widehat{\boldsymbol{U}} = \boldsymbol{X}_c - \widehat{\boldsymbol{F}}\widehat{\boldsymbol{B}}^\top$ proves both identities in (5.3).

E.2 Proximal optimality and the three-point inequality

For $\boldsymbol{y} = T_t(\boldsymbol{x})$, strict convexity of the proximal quadratic ensures a unique candidate and gives

$$(\boldsymbol{x} - \boldsymbol{y})/t - \nabla f_h(\boldsymbol{x}) \in \partial g_\lambda(\boldsymbol{y}). \tag{E.1}$$

Separability produces the identity update on the unpenalized block and soft thresholding on the penalized block in (5.6). For any finite $\boldsymbol{u}$, the subgradient inequality, convexity of f_h, and the accepted line search give

$$\begin{aligned} Q(\boldsymbol{y}) - Q(\boldsymbol{u}) &\leq \nabla f_h(\boldsymbol{x})^\top(\boldsymbol{y} - \boldsymbol{u}) + \frac{\omega}{2t}\|\boldsymbol{y} - \boldsymbol{x}\|_2^2 + g_\lambda(\boldsymbol{y}) - g_\lambda(\boldsymbol{u}) \\ &\leq \frac{(\boldsymbol{x} - \boldsymbol{y})^\top(\boldsymbol{y} - \boldsymbol{u})}{t} + \frac{\omega}{2t}\|\boldsymbol{y} - \boldsymbol{x}\|_2^2. \end{aligned}$$

Using $2(\boldsymbol{x} - \boldsymbol{y})^\top(\boldsymbol{y} - \boldsymbol{u}) = \|\boldsymbol{x} - \boldsymbol{u}\|_2^2 - \|\boldsymbol{y} - \boldsymbol{u}\|_2^2 - \|\boldsymbol{y} - \boldsymbol{x}\|_2^2$ yields

$$Q(\boldsymbol{y}) - Q(\boldsymbol{u}) \leq \frac{\|\boldsymbol{x} - \boldsymbol{u}\|_2^2 - \|\boldsymbol{y} - \boldsymbol{u}\|_2^2}{2t} - \frac{1 - \omega}{2t}\|\boldsymbol{y} - \boldsymbol{x}\|_2^2. \tag{E.2}$$

In particular, taking $\boldsymbol{u} = \boldsymbol{x}$ proves (5.10), including its factor $2 - \omega$. This uses exact minimization of the proximal surrogate, not merely comparison of that surrogate at $\boldsymbol{x}$ and $\boldsymbol{y}$.

E.3 Proof of Theorem 5.1

The global smoothness bound gives

$$f_h(\boldsymbol{y}) \leq f_h(\boldsymbol{x}) + \nabla f_h(\boldsymbol{x})^\top(\boldsymbol{y} - \boldsymbol{x}) + \frac{\mathscr{B}_h}{2}\|\boldsymbol{y} - \boldsymbol{x}\|_2^2.$$

Hence the line search must accept whenever $t \le \omega/\mathscr{B}_h$ if $\mathscr{B}_h > 0$. Starting from $t_{\max}$ and multiplying by ζ therefore takes finitely many trials. If a reduction occurs before acceptance, the previously rejected step exceeds $\omega/\mathscr{B}_h$, proving the lower bound (5.9); acceptance at the first trial gives $t_r = t_{\max}$. If $\mathscr{B}_h = 0$, the smooth objective is affine and the first trial is accepted. For this particular nonnegative pairwise loss it is constant, but affineness is already sufficient for acceptance.

Fix $\boldsymbol{x}^\star \in \mathcal{S}$. Applying (E.2) with $\boldsymbol{u} = \boldsymbol{x}^\star$ gives

$$\begin{aligned} 2t_r\{Q(\boldsymbol{x}^{(r+1)}) - Q(\boldsymbol{x}^\star)\} \le \|\boldsymbol{x}^{(r)} - \boldsymbol{x}^\star\|_2^2 - \|\boldsymbol{x}^{(r+1)} - \boldsymbol{x}^\star\|_2^2 \\ - (1-\omega)\|\boldsymbol{x}^{(r+1)} - \boldsymbol{x}^{(r)}\|_2^2. \end{aligned} \tag{E.3}$$

The left side is nonnegative, so distances to every minimizer are nonincreasing and the sequence is bounded. Summing this inequality for $r = 0, \dots, R-1$ and using monotonicity of Q yields

$$2Rt_{\min}\{Q(\boldsymbol{x}^{(R)}) - Q(\boldsymbol{x}^\star)\} \le \|\boldsymbol{x}^{(0)} - \boldsymbol{x}^\star\|_2^2,$$

which proves (5.11). Summing the descent inequality and using $t_r \le t_{\max}$ and $Q \ge 0$ gives

$$\sum_{r=0}^{\infty} \|\boldsymbol{x}^{(r+1)} - \boldsymbol{x}^{(r)}\|_2^2 < \infty.$$

Since $t_r \ge t_{\min} > 0$, $G_{t_r}(\boldsymbol{x}^{(r)}) \to 0$. By (E.1), the vectors

$$s_{r+1} = G_{t_r}(\boldsymbol{x}^{(r)}) + \nabla f_h(\boldsymbol{x}^{(r+1)}) - \nabla f_h(\boldsymbol{x}^{(r)}) \in \partial Q(\boldsymbol{x}^{(r+1)})$$

converge to zero, using global Lipschitz continuity. A convergent subsequence exists by boundedness. If its limit is $\bar{\boldsymbol{x}}$, continuity of Q and the subgradient inequality imply, for every finite $\boldsymbol{u}$,

$$Q(\boldsymbol{u}) \ge Q(\boldsymbol{x}^{(r_j)}) + s_{r_j}^\top(\boldsymbol{u} - \boldsymbol{x}^{(r_j)}) \longrightarrow Q(\bar{\boldsymbol{x}}).$$

Thus $\bar{\boldsymbol{x}} \in \mathcal{S}$. Distances to this particular minimizer are nonincreasing and have a subsequence tending to zero, so the whole sequence converges to $\bar{\boldsymbol{x}}$.

E.4 Optimality residuals and proof of Corollary 5.1

For a nonzero β_j, the jth subgradient coordinate is the singleton $g_{\beta,j} + \lambda\,\mathrm{sign}(\beta_j)$; for $\beta_j = 0$, it is the interval $[g_{\beta,j} - \lambda, g_{\beta,j} + \lambda]$.

The nuisance subgradient is always $\boldsymbol{g}_a$. Minimizing the absolute value coordinatewise proves (5.12), and the minimum is attained. Exact zero of this residual is equivalent to convex optimality. The fixed-point equivalence for G_t follows from (E.1). Adding $\nabla f_h(\boldsymbol{y})$ there proves membership in (5.14); its norm bound follows by the triangle inequality and $\|\boldsymbol{x}-\boldsymbol{y}\|_2 = t\|G_t(\boldsymbol{x})\|_2$.

We next make explicit the Section 4 event used for the finite-tolerance transfer. Put $t_\alpha = \{\log(2P/\alpha)/n\}^{1/2}$ and $\mu = 2V_n L_\varepsilon h\sqrt{2/\pi}$. The tuning condition is $\lambda \geq 4\mu + 384 V_n t_\alpha$. On the score event of that section, the slightly sharper bound

$$\|\nabla f_h(\boldsymbol{\vartheta}^\dagger)\|_\infty \leq \mu + \sqrt{8}\, V_n t_\alpha \leq \lambda/4 \tag{E.4}$$

therefore holds; the earlier proof only needed $\lambda/2$. On its contrast event and under its local curvature assumptions, every $\boldsymbol{u} \in \mathcal{C}(S)$ with $\|\boldsymbol{u}\|_2 = R = 8\sqrt{k}\lambda/\kappa$ satisfies

$$\begin{aligned} Q(\boldsymbol{\vartheta}^\dagger + \boldsymbol{u}) - Q(\boldsymbol{\vartheta}^\dagger) &\geq \frac{\kappa}{2}R^2 - (4\mu + 384 V_n t_\alpha + \lambda)\sqrt{k}R \\ &\geq \frac{\kappa}{2}R^2 - 2\lambda\sqrt{k}R = 2\lambda\sqrt{k}R. \end{aligned} \tag{E.5}$$

This inequality holds on the cone sphere itself; a sublevel inequality is not required to derive the displayed lower bound. The empirical contrast is controlled uniformly on the radius-$4\sqrt{k}R$ ℓ_1 ball. No stronger statistical event has been introduced.

Choose $\boldsymbol{s} \in \partial Q(\widetilde{\boldsymbol{\vartheta}})$ with $\|s\|_\infty \leq \varepsilon \leq \lambda/4$, and let $\boldsymbol{v} = \widetilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}^\dagger$. Convexity at $\widetilde{\boldsymbol{\vartheta}}$ gives the tilted inequality

$$Q(\widetilde{\boldsymbol{\vartheta}}) - Q(\boldsymbol{\vartheta}^\dagger) \leq \boldsymbol{s}^\top \boldsymbol{v} \leq \varepsilon\|\boldsymbol{v}\|_1. \tag{E.6}$$

Convexity of f_h at $\boldsymbol{\vartheta}^\dagger$ and the support of $\boldsymbol{\beta}_0$ also give

$$Q(\widetilde{\boldsymbol{\vartheta}}) - Q(\boldsymbol{\vartheta}^\dagger) \geq -\frac{\lambda}{4}\|\boldsymbol{v}\|_1 + \lambda\{\|(\boldsymbol{v}_\beta)_{S^c}\|_1 - \|(\boldsymbol{v}_\beta)_S\|_1\}.$$

Together with $\varepsilon \leq \lambda/4$, this implies

$$\|(\boldsymbol{v}_\beta)_{S^c}\|_1 \leq 3\|(\boldsymbol{v}_\beta)_S\|_1 + \|\boldsymbol{v}_a\|_1, \quad \boldsymbol{v} \in \mathcal{C}(S), \quad \|\boldsymbol{v}\|_1 \leq 4\sqrt{k}\|\boldsymbol{v}\|_2.$$

If $\|\boldsymbol{v}\|_2 \geq R$, let $\boldsymbol{u} = R\boldsymbol{v}/\|\boldsymbol{v}\|_2$. Convexity on the segment and (E.6) yield

$$Q(\boldsymbol{\vartheta}^\dagger + \boldsymbol{u}) - Q(\boldsymbol{\vartheta}^\dagger) \leq \varepsilon\|\boldsymbol{u}\|_1 \leq \lambda\sqrt{k}R,$$

contradicting (E.5). Hence $\|\boldsymbol{v}\|_2 \le R$ and $\|\boldsymbol{v}_\beta\|_1 \le 4\sqrt{k}R$. This is a pointwise deterministic implication on the original event, so it also applies to an outcome-dependent finite iterate satisfying the residual condition. The choices of K, h, λ must still satisfy the original fixed-tuning conditions; outcome-dependent iteration stopping does not justify outcome-dependent tuning.

E.5 Proof of Proposition 5.1 and the linear test

Let $\mathcal{P} = \{(i,j) : \Delta_i = 1,\ j \ne i\}$. Suppose a strict escape vector $\boldsymbol{v}$ exists. For every finite $(\boldsymbol{a}, \boldsymbol{\beta})$, the derivative of $Q(\boldsymbol{a} + t\boldsymbol{v}, \boldsymbol{\beta})$ with respect to t is

$$\frac{1}{N} \sum_{(i,j)\in\mathcal{P}} \Phi\{r_{ij}(\boldsymbol{a} + t\boldsymbol{v}, \boldsymbol{\beta})/h\}\, \boldsymbol{a}_{ij}^\top \boldsymbol{v} < 0.$$

Every summand is nonpositive and at least one is strictly negative, because the normal distribution function lies strictly between zero and one at a finite argument. The penalty does not change. Therefore no finite point is a minimizer.

Conversely, assume no strict escape vector exists. If $\mathcal{N}_A^\perp$ has positive dimension, define

$$\chi = \min_{\substack{v\in\mathcal{N}_A^\perp \\ \|v\|_2=1}} \frac{1}{N} \sum_{(i,j)\in\mathcal{P}} (\boldsymbol{a}_{ij}^\top v)_+.$$

The continuous function in this display is strictly positive on its compact domain. Indeed, a zero value would give $\boldsymbol{a}_{ij}^\top v \le 0$ for all pairs. The absence of strict escape would then force all products to be zero, so $v \in \mathcal{N}_A$, a contradiction. Thus $\chi > 0$.

Project $\boldsymbol{a}$ onto $\mathcal{N}_A^\perp$, writing $\boldsymbol{a}_\perp$ for this component. All pair predictions, and hence Q, are unchanged. On a nonempty sublevel set $Q \le c$, nonnegativity of ℓ_h implies $\|\boldsymbol{\beta}\|_1 \le c/\lambda$. Since $\ell_h(x+z) \ge (x+z)_+ \ge x_+ - |z|$,

$$f_h(\boldsymbol{a}, \boldsymbol{\beta}) \ge \chi \|\boldsymbol{a}_\perp\|_2 - C_Y - C_U \|\boldsymbol{\beta}\|_1,$$

where

$$C_Y = \frac{1}{N} \sum_{(i,j)\in\mathcal{P}} |Y_j - Y_i|, \qquad C_U = \frac{1}{N} \sum_{(i,j)\in\mathcal{P}} \|\widehat{u}_i - \widehat{u}_j\|_\infty$$

are finite. It follows that $\|\boldsymbol{a}_\perp\|_2 \le (c+C_Y+C_U c/\lambda)/\chi$. The projected sublevel set is closed and bounded, and thus compact. If $\mathcal{N}_A^\perp$ has dimension zero, $\boldsymbol{a}_\perp = 0$ and only the already bounded $\boldsymbol{\beta}$ block remains. Minimizing on the sublevel set through the origin gives a global minimizer in both cases. This covers $q = 0$ as well as an empty event-pair set. Flat directions may produce an unbounded set of equivalent minimizers in the original coordinates; they do not obstruct existence.

For $m > 0$, comparisons between two event patients are present in both directions. Condition (5.17) therefore forces their nuisance scores along v to be equal. Choose an event anchor i_0. All remaining conditions say that censored scores are at least the event score, with at least one strict inequality. The sum of these nonnegative margins is strictly positive and can be scaled to at least one. This proves equivalence with (5.18). In the flat case all sample nuisance scores agree; since $\widehat{\boldsymbol{A}}$ is centered, that common score is zero. Thus, for $m > 0$, $\mathcal{N}_A = \ker(\widehat{\boldsymbol{A}})$.

Finite attainment and approximate stationarity are separated by the following example. Take $n = 2$, nuisance rows $-1, 1$, $\Delta = (1, 0)^\top$, and no effective penalized predictor. The objective is $Q(a) = \ell_h(y - 2a)/2$. It has no finite minimizer, but $|Q'(a)| = \Phi\{(y - 2a)/h\} \to 0$ as $a \to \infty$. Any positive gradient tolerance will eventually be met along this escaping sequence. This is why a small residual is not, by itself, evidence of finite attainment outside the statistical event of Section 4.

E.6 Null-fit threshold and stable tail formula

If $\boldsymbol{a}^{(0)}$ is a finite nuisance-only minimizer, differentiability and convexity imply $\nabla_a f_h(\boldsymbol{a}^{(0)}, 0) = 0$. The penalized-block KKT inequality is precisely $\|\nabla_\beta f_h(\boldsymbol{a}^{(0)}, 0)\|_\infty \le \lambda$. This proves (5.19). An approximate nuisance-only fit supplies only approximate KKT conditions, which must be reported rather than treated as exact.

For $z \ge 0$, direct integration and the substitution $t = z + u$ give

$$\phi(z) - z\Phi(-z) = \int_z^\infty (t - z)\phi(t)\,\mathrm{d}t = \phi(z) \int_0^\infty u \exp(-zu - u^2/2)\,\mathrm{d}u.$$

Since $\ell_h(r) - r_+$ is even, this proves (5.20), including $J(0) = 1$. If $z > 0$, substituting $v = zu$ gives

$$J(z) = z^{-2} \int_0^\infty v\mathrm{e}^{-v} \exp\{-v^2/(2z^2)\} \, \mathrm{d}v.$$

For $x \geq 0$, Taylor's formula for e^{-x} after M terms has remainder of sign $(-1)^M$ and magnitude at most $x^M/M!$. Integrating the finite expansion gives

$$z^{-2} \int_0^\infty v\mathrm{e}^{-v} \frac{(-1)^j v^{2j}}{2^j j! z^{2j}} \, \mathrm{d}v = \frac{(-1)^j (2j+1)!}{2^j j! z^{2j+2}} = \frac{(-1)^j (2j+1)!!}{z^{2j+2}}.$$

The same bound on the integrated remainder proves (5.21). The remainder has the sign of the next term, so adjacent truncations bracket the integral in exact arithmetic. The series is asymptotic, not generally convergent for a fixed finite z; a numerical implementation must stop according to a remainder bound before terms grow. Tail underflow, if encountered, must be distinguished from an exact zero of Φ or ℓ_h in the existence and convergence arguments.

F Numerical reproducibility and work accounting

The computational illustrations use the synthetic design in Section 6.1. Each scenario has one generated training sample. The training-selection record contains all 144 fold candidates, 48 aggregate positions, and 12 selected full-training models, including numerical eligibility and grid-boundary indicators. The three endpoint corrections in Section 6.3 and the nine fold-level corrections in Section 6.4 are separate sets of Newton calls.

F.1 Data generation and training folds

The original synthetic generator uses the Mersenne–Twister, Inversion, and Rejection random-number generator (RNG) settings in R. Matrices are filled by columns, with draw order $\boldsymbol{F}, \boldsymbol{v}, \boldsymbol{Z}, \varepsilon, V$. For scenario index j and replicate $r = 1$, the seed rule is $20260929 + 100000j + 100r +$ role, with roles 1 and 2 for training and test samples. The six training samples have event counts 87, 83, 87, 89, 46, and 93 for S1–S6. Population censoring is calibrated by (6.2); sample censoring retains its generated variation.

The three-fold training-selection analysis uses the event-stratified labels stored with the experiment configuration. Both methods share these labels; each complement contains 80 patients and each validation fold 40. Every mean and PCA transformation is learned on its training complement. The aggregate score retains all within-fold ordered pairs and the common zero-prediction reference in (3.18).

Earlier evaluations of the generated 400-patient test samples form part of the development record. The cross-validation analysis in this article uses training samples throughout. Assessment of the complete selected procedure on new independent samples is a separate evaluation.

F.2 Computational work

A FISTA update is an accepted proximal update, and a Newton update is an accepted active-face correction. The training-selection run has the following work counts; initialization and endpoint verification are counted separately.

Operation	Calls	Accepted updates	Gradient evaluations
Nuisance initialization	48	3,531	3,579 plus 48 full-design
FISTA	156	840,530	1,680,940
Active-face refinement	9	9	18

Endpoint verification additionally uses 165 core and 165 independent pairwise gradient evaluations. The nine refinements require nine Hessian constructions and linear solves; all accept full steps without halving. They occur in UA fold candidates, while the twelve selected full-training refits use no Newton updates. The complete run takes 7,709.2 seconds on the recorded Apple M4 system, including preprocessing, initialization, fitting, verification, and input/output. This is a single-run resource measurement.

F.3 Source objects and environment

The reproducibility record retains all aggregate scores, candidate statuses, selected-model summaries, and final coefficient vectors. The identity map specifies the original fields and source hashes. Coefficient nonzero counts use exact comparison with zero. The final standardized objects reconstruct coefficients, the deployment slope, training scores, gradients, and KKT values from the final coefficient vector, including candidates completed by refinement.

The computation uses R 4.5.2 on Apple M4 arm64 with macOS 26.5, Apple Accelerate Basic Linear Algebra Subprograms (BLAS), and Linear Algebra Package (LAPACK) 3.12.1. The original environment includes Python 3.12.15 for independent checks. Package versions and numerical controls are recorded in the accompanying environment and configuration files. Saved R serialized-data (RDS) objects remain the numerical originals; exported tables and figures are linked to those objects. Full trajectories and checkpoints remain in the computational archive. Artwork was generated in R and reformatted by vector-path-preserving operations with PyMuPDF and Ghostscript; plotted coordinates were retained.

Figure S1 presents the accepted-update counterpart of the main-text stationarity trajectories.

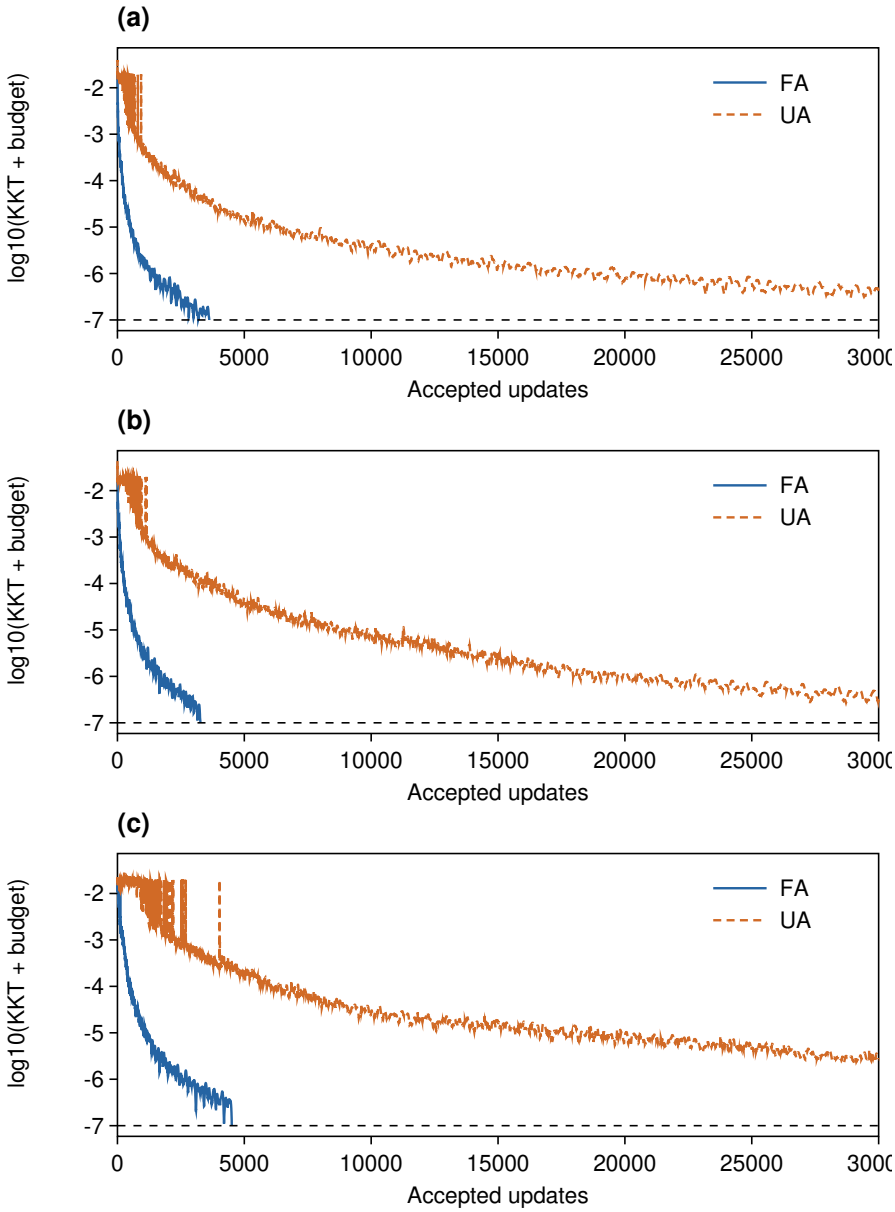


Fig. S1 Full original-coordinate KKT residuals plus tracked numerical budgets versus accepted FISTA updates. The horizontal line is 10^{-7}. Panels (a), (b), and (c) show S3, S4, and S6. FA is solid and UA is dashed. The fits are the same as Figure 1, ending before active-face refinement

G Implementation conventions and numerical diagnostics

Objective and coordinates. The implementation retains ordered off-diagonal pairs, event weights Δ_i, residual direction $e_j - e_i$, and normalization $n(n-1)$. Censored patients remain comparison subjects. Clinical and factor coefficients are unpenalized; molecular coefficients use the declared ℓ_1 penalty and original coordinate scales. For rescaled coordinates, Section 3.1 gives the equivalent weighted penalty. Covariate means and the PCA map are learned within training data and frozen for validation. The recorded observation model determines the treatment of rounded times and ties.

Finite attainment and stationarity. The complete KKT residual uses every nuisance and molecular coordinate and the exact zero/nonzero status of the stored coefficients. The nuisance-direction diagnostic is recorded separately from stationarity. Exact zero of the complete residual characterizes a finite minimizer of the convex criterion. A positive-tolerance numerical result is reported as stationarity to the declared tolerance; the archived output retains `global_optimum_certified=FALSE`.

Tracked arithmetic budget. The numerical allowance E tracks known tail and underflow contributions using normal-tail bounds and log-scale calculations. It is combined with the computed residual for numerical acceptance. It is not a complete interval enclosure of every floating-point operation. Nonfinite predictors, gradients, or losses and unresolved line searches receive distinct status codes. The expansion in (5.21) is truncated using its remainder bound before asymptotic terms grow. A positive quantity below representable range retains its logarithmic bound rather than being interpreted as a mathematical zero.

Stable validation contrasts. Validation computes the positive-part difference within each ordered pair before aggregation. With $a = Y_j - Y_i$ and $b = \widehat{\eta}_i - \widehat{\eta}_j$, this is $(a+b)_+ - a_+$. Sign-based evaluation returns b when both positive parts are active, avoiding cancellation of two large totals. The bound $|(a+b)_+ - a_+| \le |b|$ provides a deterministic arithmetic check. Comparisons are formed within each

validation fold, and the denominator is the sum of its ordered-pair counts.

Paths, refinement, and candidate eligibility. The null-fit threshold is evaluated after nuisance-only fitting and a complete-design gradient calculation, with the numerical budget propagated to its reported status. Penalty levels remain strictly positive. Warm starts advance from the last successful candidate. Active-face refinement retains the input support and signs, directly solves a numerically positive-definite Hessian system, and verifies all inactive-coordinate inequalities at the accepted point. The procedure in Section 5.9 fixes the trigger and update budget. An invalid fold makes the corresponding aggregate unavailable; all candidate positions, grid boundaries, and status reasons remain in the record.

References

Bai, J., & Ng, S. (2013). Principal components estimation and identification of static factors. *Journal of Econometrics*, *176*(1), 18–29. https://doi.org/10.1016/j.jeconom.2013.03.007

Clémençon, S., Lugosi, G., & Vayatis, N. (2008). Ranking and empirical minimization of U-statistics. *The Annals of Statistics*, *36*(2), 844–874. https://doi.org/10.1214/009052607000000910

Ledoux, M., & Talagrand, M. (1991). *Probability in Banach Spaces: Isoperimetry and Processes*. Springer. https://doi.org/10.1007/978-3-642-20212-4

Vershynin, R. (2012). Introduction to the non-asymptotic analysis of random matrices. In Y. C. Eldar, & G. Kutyniok (Eds.), *Compressed Sensing: Theory and Applications* (pp. 210–268). Cambridge University Press. https://doi.org/10.1017/CBO9780511794308.006